

SOC Architecture Project Report

1. Project Title

Cloud-Based SOC Architecture using Wazuh SIEM, Suricata IDS, ModSecurity WAF, and Shuffle SOAR on AWS

2. Project Overview

This project demonstrates the design and implementation of a cloud-based Security Operations Center (SOC) architecture deployed in an AWS environment using multiple EC2 instances. The architecture integrates various open-source security technologies to provide centralized monitoring, intrusion detection, web application protection, threat intelligence correlation, automated alerting, and security event analysis.

The primary objective of the project was to simulate real-world cyberattacks against a vulnerable web application and evaluate the effectiveness of the deployed security controls in detecting, monitoring, correlating, and responding to security incidents.

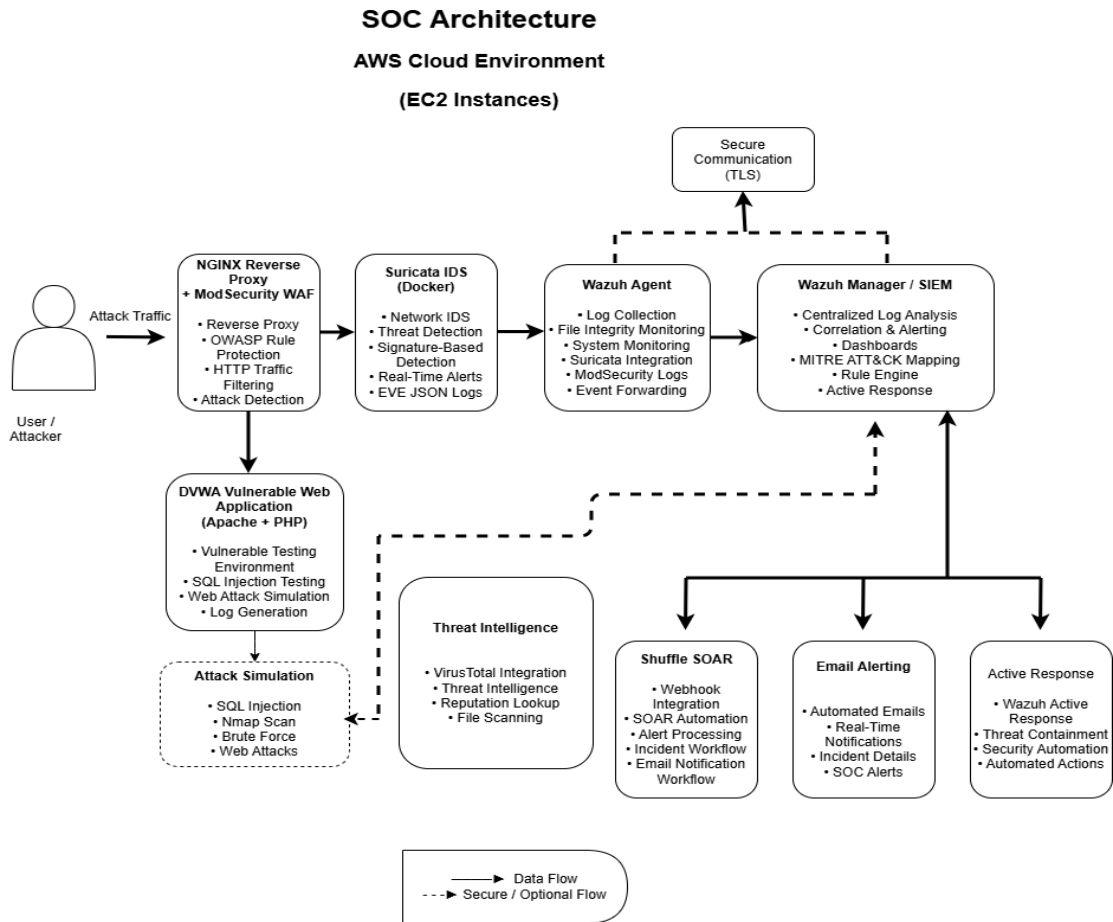
The SOC environment was built using the following technologies:

- Wazuh SIEM
- Suricata IDS
- ModSecurity WAF
- NGINX Reverse Proxy
- DVWA (Damn Vulnerable Web Application)
- Shuffle SOAR
- VirusTotal Threat Intelligence
- Email Alerting Automation

The project successfully demonstrated:

- Centralized log collection
- Real-time attack detection
- Web application firewall protection
- Intrusion detection
- Threat intelligence monitoring
- Automated security alerting
- SOC event correlation
- Attack simulation and monitoring

3. SOC Architecture Diagram



4. AWS Infrastructure Setup

The SOC architecture was deployed inside an AWS cloud environment using EC2 instances.

Infrastructure Details

Component	Details
Cloud Provider	AWS
VPC	Default AWS VPC
EC2 Instances	2 Instances
Operating System	Ubuntu Server 22.04
Web Server	NGINX

Vulnerable Application	DVWA
SIEM Platform	Wazuh
IDS	Suricata
SOAR Platform	Shuffle
Threat Intelligence	VirusTotal
Notification System	Email Alerting

5. Architecture Components

5.1 NGINX Reverse Proxy + ModSecurity WAF

The NGINX server acted as the public-facing reverse proxy server responsible for handling incoming traffic from users and attackers.

ModSecurity WAF with OWASP Core Rule Set (CRS) was integrated with NGINX to inspect incoming HTTP requests and block malicious payloads.

Functions

- Reverse proxy functionality
- HTTP traffic filtering
- Web application firewall protection
- OWASP CRS rule enforcement
- Detection of malicious payloads
- Blocking suspicious requests

Attacks Detected

- SQL Injection
- Command Injection
- Malicious HTTP Requests
- File Upload Abuse
- Suspicious URL Patterns

[Insert Screenshot: NGINX + ModSecurity Configuration]

5.2 DVWA Vulnerable Web Application

DVWA (Damn Vulnerable Web Application) was deployed for controlled attack simulation and testing purposes.

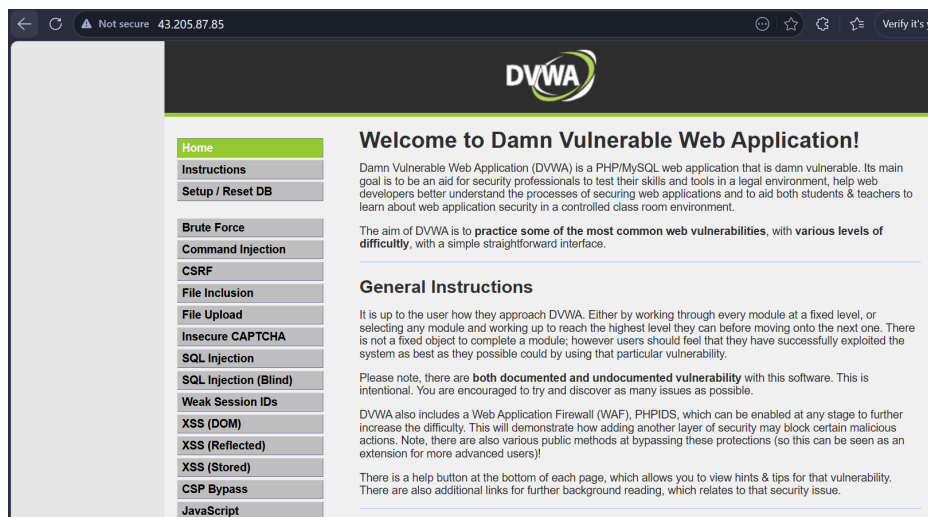
The application was intentionally vulnerable to common web attacks and used to validate detection and monitoring capabilities of the SOC environment.

Functions

- Attack simulation environment
- Web vulnerability testing
- Log generation
- Security validation

Vulnerabilities Tested

- SQL Injection
- Command Injection
- File Upload Vulnerability
- Reconnaissance Scanning



5.3 Suricata IDS

Suricata was configured as a Network Intrusion Detection System (NIDS) for monitoring network traffic and generating alerts based on malicious activity signatures.

Suricata generated alerts using Emerging Threats (ET) rules and forwarded logs to Wazuh SIEM through EVE JSON logging.

Functions

- Real-time network traffic monitoring

- Signature-based attack detection
- Threat intelligence monitoring
- HTTP traffic inspection
- Alert generation

Detection Capabilities

- Reconnaissance detection
- HTTP attack detection
- Threat intelligence alerts
- Reputation-based IP monitoring
- Web attack detection

Logging

Suricata EVE JSON logging was enabled and integrated with Wazuh.

[Insert Screenshot: Suricata Configuration]

[Insert Screenshot: Suricata Alerts in Wazuh Dashboard]

5.4 Wazuh SIEM

Wazuh served as the centralized SIEM platform for collecting, analyzing, and correlating security logs from integrated security tools.

The Wazuh Manager received logs from:

- Suricata IDS
- NGINX
- ModSecurity
- Wazuh Agent

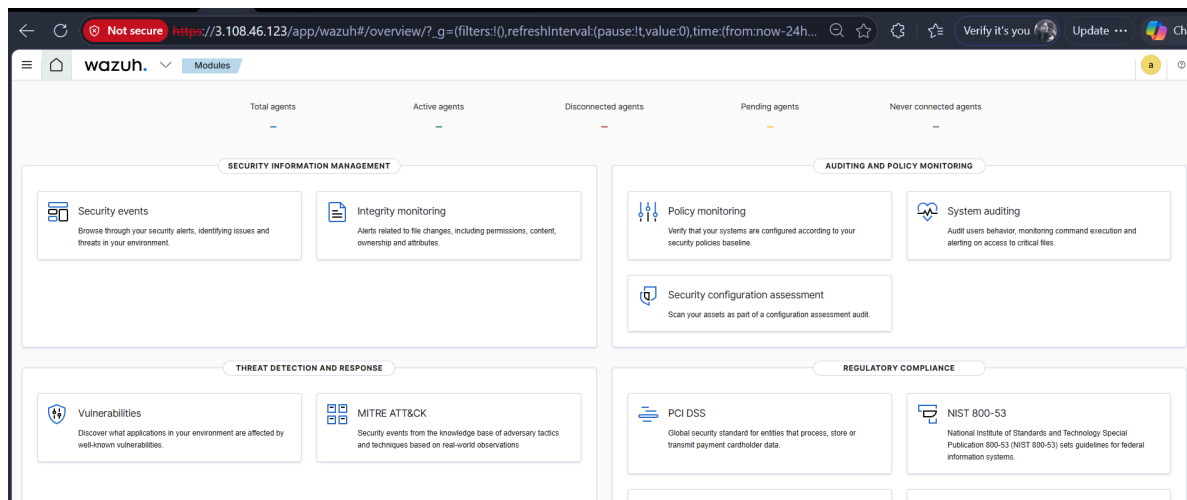
Functions

- Centralized log collection
- Event correlation
- Alert generation
- Security dashboards
- MITRE ATT&CK mapping
- Security monitoring

Key Features Implemented

- Real-time event monitoring
- Security alert correlation

- Attack visibility
- Threat monitoring
- Log management



5.5 Shuffle SOAR Integration

Shuffle SOAR was integrated for automated incident notification and workflow automation.

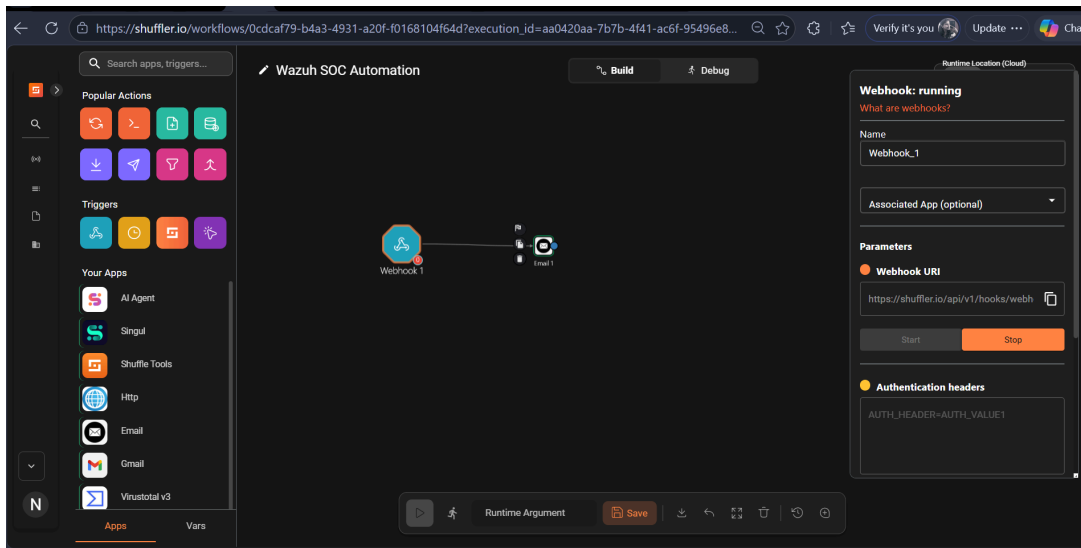
Webhook integration was configured between Wazuh and Shuffle to automate alert forwarding.

Functions

- Webhook automation
- Security alert forwarding
- Email notification automation
- Incident workflow automation

Workflow

1. Wazuh generates alert
2. Alert forwarded to Shuffle webhook
3. Shuffle processes alert
4. Email notification generated



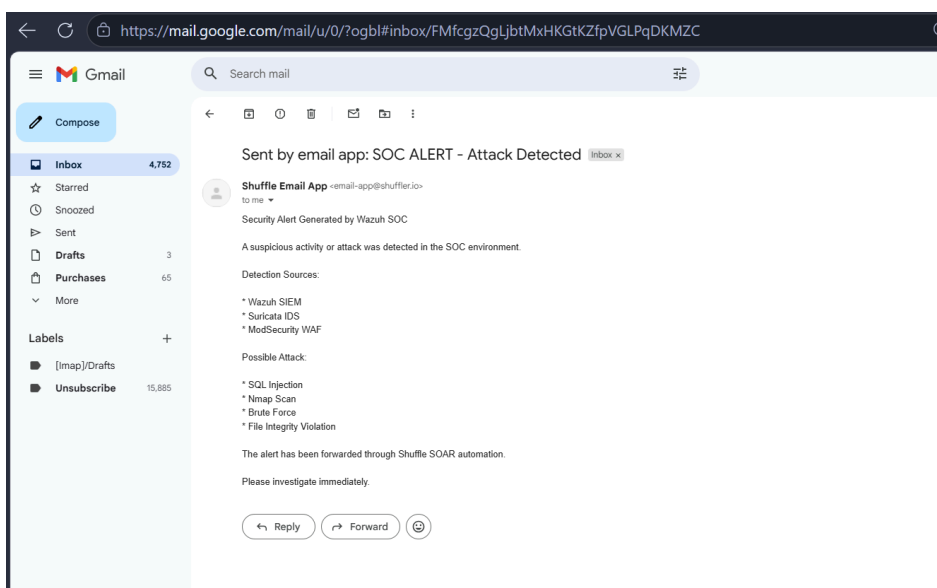
5.6 Email Alerting

Automated email notifications were configured to provide real-time SOC alerts.

Security incidents detected by Wazuh were automatically forwarded through Shuffle SOAR email workflows.

Email Contents

- Alert description
- Attack information
- Detection source
- Timestamp
- Security context



6. Attack Simulations Performed

Several attack simulations were performed against the DVWA application to validate the effectiveness of the SOC architecture.

6.1 SQL Injection Attack Simulation

SQL Injection payloads were tested against the DVWA SQL Injection module.

Objective

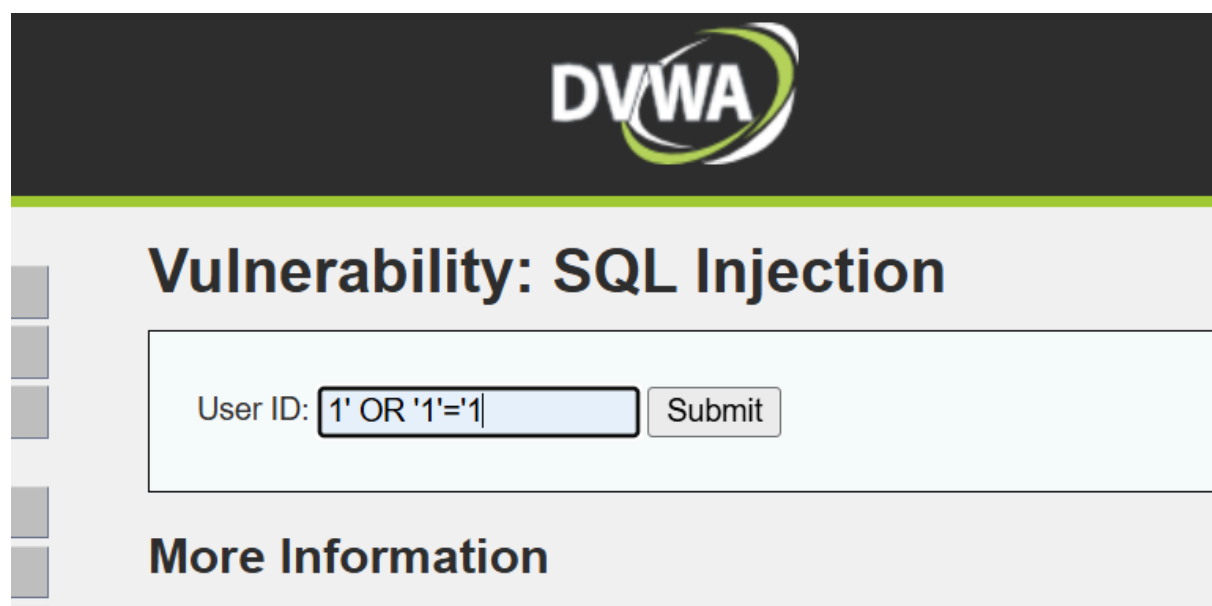
To verify whether malicious SQL payloads could be detected and blocked by ModSecurity and monitored by Suricata and Wazuh.

Results

- ModSecurity detected malicious SQL payloads
- HTTP 403 Forbidden responses generated
- Suricata generated IDS alerts
- Wazuh correlated and displayed alerts
- Email notifications generated

Detection Sources

- ModSecurity WAF
- Suricata IDS
- Wazuh SIEM



403 Forbidden

nginx/1.24.0 (Ubuntu)

Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> May 12, 2026 @ 21:46:20.343	001	ip-172-31-35-150	T1083	Discovery	ModSecurity rejected a query	7	31333
> May 12, 2026 @ 21:46:20.302	001	ip-172-31-35-150			Web server 400 error code.	5	31101

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Shuffle Email App <email-app@shuffler.io>
to me ▾

20:13 (2 minutes ago)



Shuffle Email App <email-app@shuffler.io>
to me ▾

20:14 (1 minute ago)

Security Alert Generated by Wazuh SOC

A suspicious activity or attack was detected in the SOC environment.

Detection Sources:

- * Wazuh SIEM
- * Suricata IDS
- * ModSecurity WAF

Possible Attack:

- * SQL Injection
- * Nmap Scan
- * Brute Force
- * File Integrity Violation

The alert has been forwarded through Shuffle SOAR automation.

Please investigate immediately.

↩ Reply

➡ Forward

🗑

6.2 Command Injection Attack Simulation

Command injection payloads were tested using the DVWA command execution module.

Example payload used:

127.0.0.1; whoami

Objective

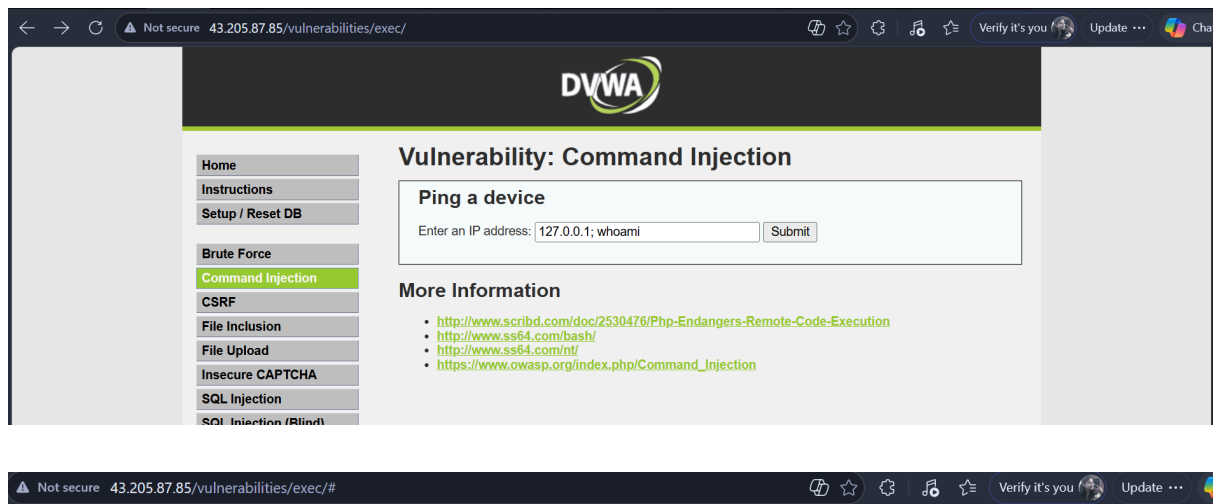
To test detection of malicious command execution attempts.

Results

- Request blocked by ModSecurity
- HTTP 403 Forbidden generated
- Suricata detected malicious activity
- Wazuh displayed correlated alerts

Detection Sources

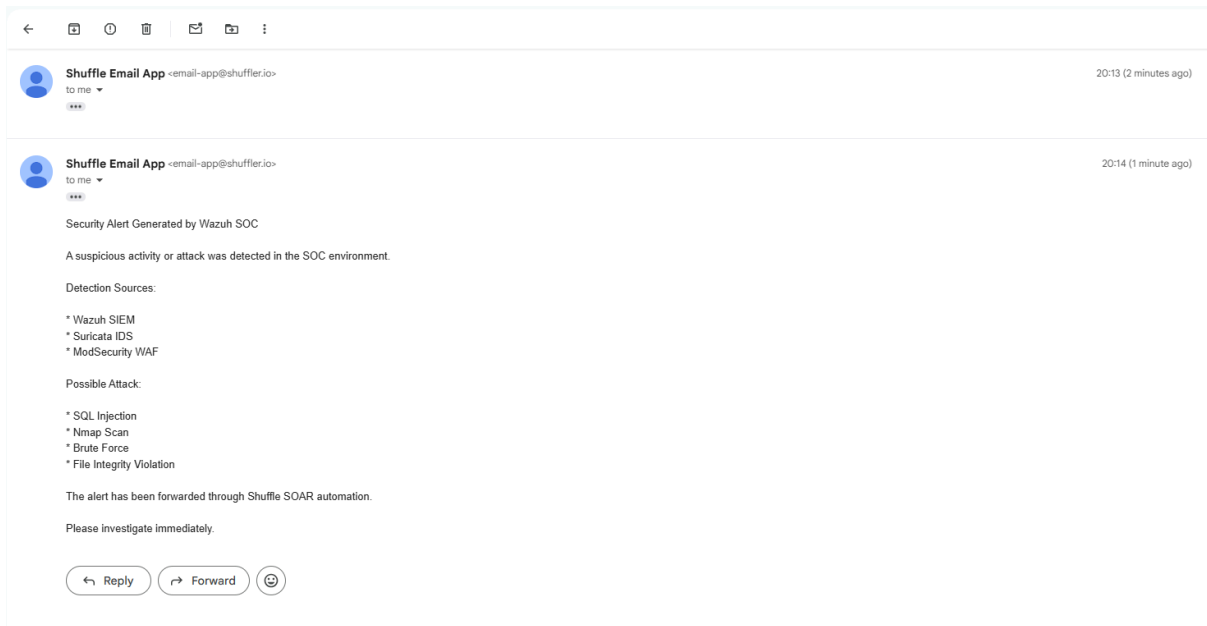
- ModSecurity WAF
- Suricata IDS
- Wazuh SIEM



403 Forbidden

nginx/1.24.0 (Ubuntu)

Security Alerts								
	Time ↕	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
✓	May 12, 2026 @ 21:47:00.345	001	ip-172-31-35-150			Suricata: Alert - GPL_WEB_SERVER 403 Forbidden	3	86601
TableJSONRule								
	@timestamp		2026-05-12T16:17:06.345Z					
	._id		IWF6H-U4Bua0rwZAYaPgF					
	agent.id		001					
	agent.ip		172.31.35.150					
	agent.name		ip-172-31-35-150					
	data.alert.action		allowed					
	data.alert.category		Attempted Information Leak					
	data.alert.gid		1					
	data.alert.metadata.created_at		2010_09_23					
	data.alert.metadata.signature_severity		Unknown					
	data.alert.metadata.updated_at		2024_03_08					



6.3 File Upload Vulnerability Testing

A file upload attack simulation was performed using a PHP shell payload ([shell.php](#)) against the DVWA file upload functionality.

Objective

To test unrestricted file upload detection and monitoring capabilities.

Results

- Malicious upload attempt monitored
- Security events generated
- Alerts centralized in Wazuh
- Activity visible in SOC dashboard

Detection Sources

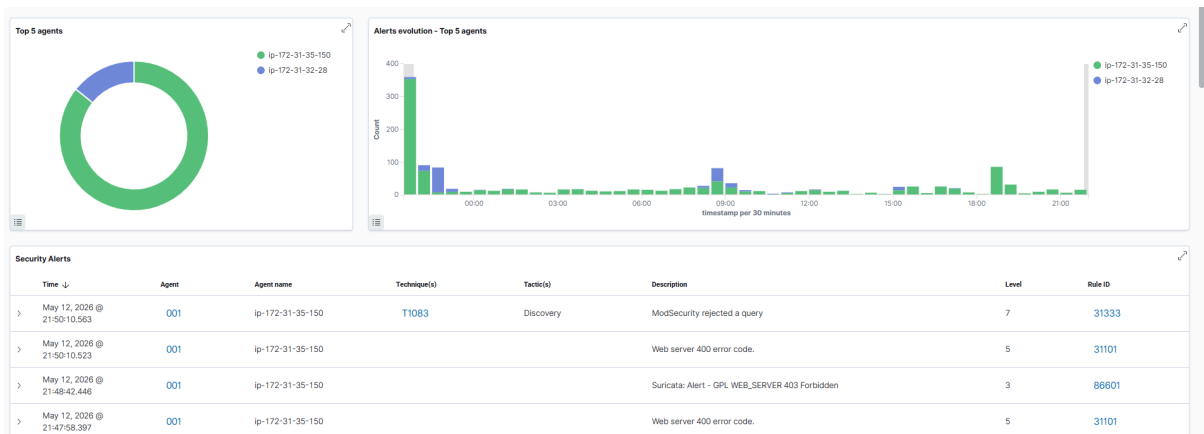
- ModSecurity WAF
- Wazuh SIEM

Vulnerability: File Upload

Choose an image to upload:

Choose File shell.php

Upload



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Shuffle Email App <email-app@shuffler.io> 20:13 (2 minutes ago)
to me

Shuffle Email App <email-app@shuffler.io> 20:14 (1 minute ago)
to me

Security Alert Generated by Wazuh SOC

A suspicious activity or attack was detected in the SOC environment.

Detection Sources:

- * Wazuh SIEM
- * Suricata IDS
- * ModSecurity WAF

Possible Attack:

- * SQL Injection
- * Nmap Scan
- * Brute Force
- * File Integrity Violation

The alert has been forwarded through Shuffle SOAR automation.

Please investigate immediately.

↩ Reply ↪ Forward 😊

6.4 Reconnaissance and Nmap Scanning

Nmap scanning was performed against the public-facing server to simulate reconnaissance activity.

Objective

To evaluate IDS detection of suspicious scanning activity.

Results

- Reconnaissance activity detected
- Suricata generated alerts
- Events forwarded to Wazuh

Detection Sources

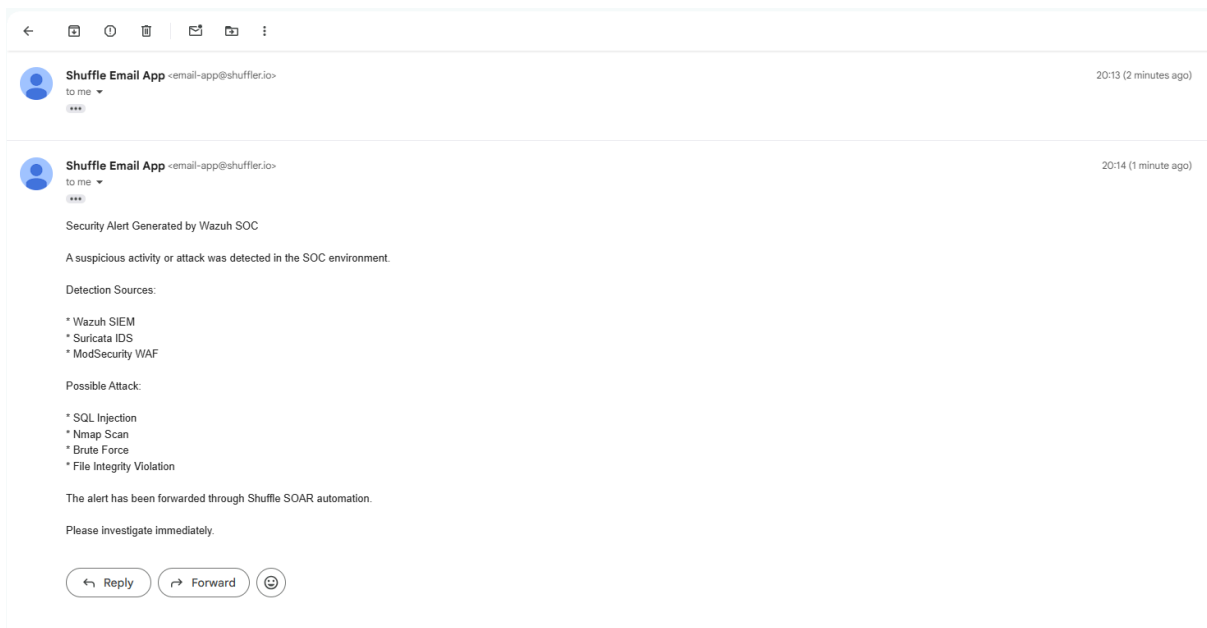
- Suricata IDS
- Wazuh SIEM

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 22:02 +0530
Nmap scan report for ec2-43-205-87-85.ap-south-1.compute.amazonaws.com (43.205.87.85)
Host is up (0.034s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     nginx 1.24.0 (Ubuntu)
|_http-title: 403 Forbidden
443/tcp   closed https
Device type: general purpose
Running (JUST GUESSING): Linux 2.6.X (86%)
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
Aggressive OS guesses: Linux 2.6.32 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 17 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 443/tcp)
HOP RTT      ADDRESS
1   3.00 ms  reliance.reliance (192.168.29.1)
2   5.00 ms  10.28.72.1
3   6.00 ms  172.31.1.104
4   5.00 ms  192.168.242.32
5   6.00 ms  192.168.231.4
6   7.00 ms  192.168.231.19
7   8.00 ms  192.168.64.4
8   ... 16
17  32.00 ms ec2-43-205-87-85.ap-south-1.compute.amazonaws.com (43.205.87.85)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 149.00 seconds
```

Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
May 12, 2026 @ 22:05:33.668	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651
May 12, 2026 @ 22:05:33.473	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651
May 12, 2026 @ 22:05:33.422	001	ip-172-31-35-150	T1046	Discovery	Nmap Scan Detected by Suricata	12	100200
May 12, 2026 @ 22:05:28.419	001	ip-172-31-35-150			Suricata: Alert - ET CINS Active Threat Intelligence Poor Reputation IP group 153	3	86601
May 12, 2026 @ 22:04:25.359	001	ip-172-31-35-150			Suricata: Alert - ET SCAN Potential SSH Scan	3	86601
May 12, 2026 @ 22:03:37.322	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651
May 12, 2026 @ 22:03:37.315	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651
May 12, 2026 @ 22:03:35.460	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651
May 12, 2026 @ 22:03:35.460	001	ip-172-31-35-150	T1046	Discovery	Nmap Scan Detected by Suricata	12	100200
May 12, 2026 @ 22:03:35.460	001	ip-172-31-35-150			Host Blocked by firewall-drop Active Response	3	651



7. Threat Intelligence Monitoring

The SOC architecture successfully detected traffic originating from known malicious and poor-reputation IP addresses using Emerging Threats (ET) intelligence feeds integrated within Suricata IDS.

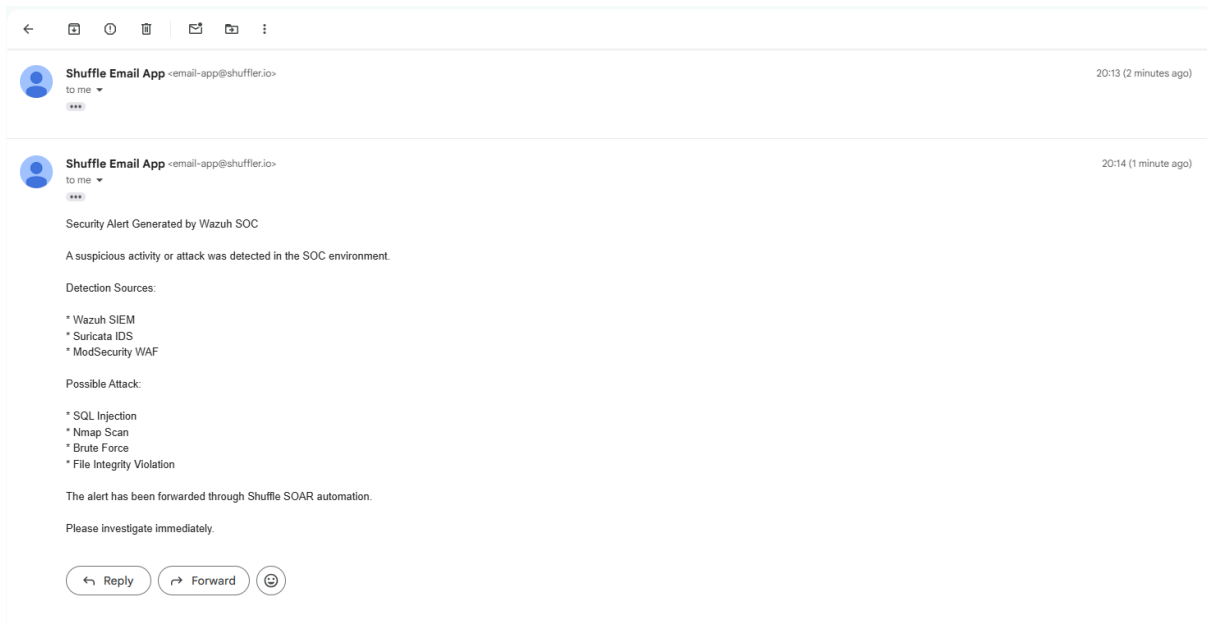
Alerts Detected

- ET DROP Dshield Block Listed Source
- ET CINS Active Threat Intelligence Poor Reputation IP

Results

- Real-world malicious IP activity detected
- Reputation-based detection successful
- Alerts centralized in Wazuh SIEM
- Threat visibility improved





8. Security Measures Implemented

The following preventive and detective security controls were implemented:

Security Control	Purpose
ModSecurity WAF	Web application protection
OWASP CRS Rules	Attack signature filtering
Suricata IDS	Network intrusion detection
Wazuh SIEM	Centralized monitoring
Shuffle SOAR	Automation and alerting
Email Notification	Incident response
Threat Intelligence Rules	Reputation-based detection

9. Findings and Observations

The project successfully demonstrated that:

- Web attacks can be detected in real-time
- Centralized SIEM monitoring improves visibility
- WAF protection effectively blocks malicious payloads
- IDS systems provide network-level monitoring

- SOAR automation improves incident response efficiency
- Threat intelligence feeds enhance detection capability

The integration between Wazuh, Suricata, ModSecurity, and Shuffle SOAR successfully created an automated SOC monitoring environment.

10. Improvements Implemented

Several improvements were added during implementation:

- Reduced unnecessary Suricata logging
- Optimized EVE JSON configuration
- Improved Wazuh log parsing
- Enabled centralized alert correlation
- Added SOAR webhook integration
- Added automated email notifications
- Integrated threat intelligence detection

11. Conclusion

This project successfully implemented a cloud-based SOC architecture capable of:

- Detecting web application attacks
- Monitoring suspicious network activity
- Correlating security events
- Generating centralized alerts
- Automating notifications
- Integrating threat intelligence feeds

The combination of Wazuh SIEM, Suricata IDS, ModSecurity WAF, Shuffle SOAR, and DVWA created a realistic SOC environment for attack simulation, monitoring, and incident detection.

The project demonstrates how open-source security tools can be integrated to build an effective SOC monitoring and response platform in a cloud environment.